

KENLEY AF, United Kingdom

WMO: 037810

Lat: 51.3039N

Lon: 0.0914W

Elev: 170

StdP: 99.30

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-2.0	-6.1	2.3	0.2	-4.7	2.6	0.5	12.8	9.2	11.7	9.0	2.7	30	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	26.6	18.3	24.5	17.3	22.7	16.6	19.2	24.5	18.2	22.8	17.4	21.3	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.7	20.7	16.6	12.0	19.9	15.8	11.4	19.1	55.3	24.6	52.1	22.9	49.5	21.3	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.1	8.1	DB	-5.3	29.7	1.9	2.3	-6.7	31.4	-7.8	32.7	-8.9	34.0	-10.3	35.7	(4)
(5)			WB	-5.5	21.3	1.9	1.7	-6.9	22.5	-8.0	23.5	-9.1	24.5	-10.5	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.5	4.8	6.5	9.0	11.9	14.9	17.2	16.9	14.4	11.1	7.5	5.2		
	DBStd	5.41	3.18	3.22	3.05	3.14	3.06	2.67	2.58	2.55	2.49	2.78	3.10	3.40		
	HDD10.0	753	172	147	112	53	14	1	0	0	1	19	84	151		
	HDD18.3	2952	429	379	366	279	200	108	54	60	122	223	324	407		
	CDD10.0	887	1	1	4	24	73	148	223	214	132	54	10	2		
	CDD18.3	44	0	0	0	0	1	6	19	16	3	1	0	0		
	CDH23.3	297	0	0	0	1	6	42	129	104	13	1	0	0		
	CDH26.7	48	0	0	0	0	0	6	21	21	1	0	0	0		
(14) Wind	WSAvg	3.9	4.6	4.5	4.1	3.8	3.7	3.6	3.5	3.4	3.4	3.8	4.0	4.3		
(15) Precipitation	PrecAvg	708	71	50	42	49	49	52	46	58	57	75	78	70		
	PrecMax	994	186	117	105	130	107	136	104	108	123	173	166	152		
	PrecMin	501	12	6	10	3	7	9	9	2	12	21	33	10		
	PrecStd	126	37	31	23	32	25	36	24	28	31	38	38	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.6	16.6	22.0	24.2	27.3	29.0	29.0	25.0	20.0	15.1	12.9		
		MCWB	10.5	9.8	10.9	13.9	16.0	18.6	19.2	19.3	17.5	15.8	13.7	11.7		
	2%	DB	11.1	11.3	14.1	18.4	21.7	24.2	26.7	26.4	22.2	17.3	13.9	12.1		
		MCWB	9.9	9.4	9.9	12.2	15.2	17.0	18.3	18.4	16.8	14.5	12.6	11.1		
	5%	DB	10.2	10.2	12.4	16.0	19.6	22.1	24.6	23.7	20.1	16.1	12.9	11.2		
		MCWB	9.1	8.5	9.2	10.8	14.2	16.1	17.5	17.3	15.7	14.0	11.6	10.2		
	10%	DB	9.2	9.2	11.0	14.1	17.4	20.2	22.6	21.6	18.5	15.1	12.0	10.3		
		MCWB	8.2	7.8	8.6	9.8	12.8	15.1	16.6	16.4	14.8	13.5	10.7	9.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.9	11.7	14.7	17.7	19.7	20.7	20.6	19.0	16.6	14.2	12.1		
		MCDB	11.6	12.2	14.5	20.7	22.0	25.8	27.2	27.1	22.9	18.9	14.6	12.6		
	2%	WB	10.3	9.9	10.7	12.8	16.1	18.0	19.2	19.2	17.6	15.4	13.0	11.3		
		MCDB	10.9	11.0	12.8	17.4	20.2	22.5	24.7	24.4	20.9	16.7	13.7	11.9		
	5%	WB	9.2	8.9	9.9	11.5	14.7	16.8	18.2	18.1	16.6	14.6	11.9	10.5		
		MCDB	10.0	9.9	11.8	14.8	18.6	20.8	23.1	22.1	19.1	15.7	12.8	11.2		
	10%	WB	8.3	8.0	9.0	10.5	13.4	15.7	17.2	17.2	15.7	13.8	10.9	9.4		
		MCDB	9.1	9.0	10.8	13.3	16.8	19.4	21.4	20.6	17.7	14.9	11.9	10.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.0	6.2	7.6	7.9	8.1	8.3	7.9	7.3	5.9	4.9	4.4		
		MCDBR	5.2	5.7	8.3	10.2	11.0	11.1	11.6	10.8	9.3	6.4	4.8	4.5		
	5% WB	MCWBR	5.3	4.4	5.2	5.6	6.0	5.8	5.2	4.9	4.9	4.4	4.3	4.6		
		MCWBR	5.2	5.3	7.1	9.1	10.0	10.1	9.3	8.0	5.7	4.7	4.7	4.5		
(40) Clear-Sky Solar Irradiance	taub	0.324	0.340	0.375	0.394	0.399	0.399	0.399	0.395	0.387	0.370	0.354	0.328	0.314		
	taud	2.388	2.395	2.289	2.256	2.283	2.305	2.321	2.364	2.409	2.438	2.432	2.432	2.373		
	Ebn at Noon	688	783	815	843	856	857	854	842	816	756	680	634	54		
	Edh at Noon	62	80	106	123	125	124	120	109	94	76	59	54	54		
(44) All-Sky Solar Radiation	RadAvg	0.81	1.48	2.61	4.11	4.94	5.47	5.20	4.33	3.24	1.86	0.97	0.63			
	RadStd	0.07	0.20	0.34	0.42	0.33	0.50	0.47	0.34	0.26	0.17	0.11	0.08			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page